BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 136 · Network Performance

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is FTP used for?

Q2. How does phishing trick people?
A. Tries every possible password
B. Floods a server with traffic
C. Repairs damaged files
D. Pretends to be trusted

Q3. “Bluetooth is normally the best choice for a high-capacity link between distant school sites.” Correct this. Explain why.

Q4. A network shows 2001:db8::7. “IPv6 uses four decimal numbers separated by dots.” Correct this. How is this address written?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Bandwidth — The available data transfer capacity of a connection over time.

Latency — Delay before data begins or a response is received.

Wired and wireless — Wired links tend to be less affected by wireless interference; wireless provides mobility.

Performance factors — Traffic, distance, medium, devices, interference and service contention affect observed speed.

New learning

Bandwidth: The available data transfer capacity of a connection over time.

Latency: Delay before data begins or a response is received.

Wired and wireless: Wired links tend to be less affected by wireless interference; wireless provides mobility.

Performance factors: Traffic, distance, medium, devices, interference and service contention affect observed speed.

Worked example

A 24 Mb file travels at a constant 8 Mb/s. Calculate ideal transfer time.

1. Identify units: both are megabits.

2. Use time = file size ÷ transfer rate.

3. Compute 24 ÷ 8 = 3 seconds.

4. Explain this ideal figure excludes latency/overhead.

We Do · Practise with guidance

Guided 1

A 60 Mb file travels at 12 Mb/s. Calculate ideal time.

Worked method: 5 seconds.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Correct the calculation of transfer time in seconds.

```
sizeMb = 90
speedMbps = 30
seconds = sizeMb * speedMbps
print(seconds)
```

iterate

Question 1

Explain the difference between bandwidth and latency. [2 marks]

Question 2

Calculate time for a 40 Mb file over 10 Mb/s ideal link. [2 marks]

Question 3

Give two factors reducing Wi-Fi performance. [2 marks]

Exit Ticket · Answer before checking

Bandwidth and latency are exactly the same measurement.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Transferring files between networked computers.

Q2. D — Pretends to be trusted

Q3. Bluetooth suits short-range links, such as keyboards or headphones. Check range, data rate and power needs.

Q4. IPv6 uses hexadecimal groups separated by colons. Groups of zeros can be shortened. Routers use IP addresses between networks; switches use MAC addresses within a LAN.

I Do / We Do pages · Guided practice

Q1. 5 seconds.

You Do pages · Code First

```
sizeMb = 90
speedMbps = 30
seconds = sizeMb / speedMbps
print(seconds)
```

Expected result: 3

You Do pages · Independent questions

Q1. Bandwidth is data capacity per unit time; latency is delay before response/transfer.

Q2. 40 ÷ 10 = 4 seconds.

Q3. Interference, weak signal/distance, number of users or obstacles.

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: